

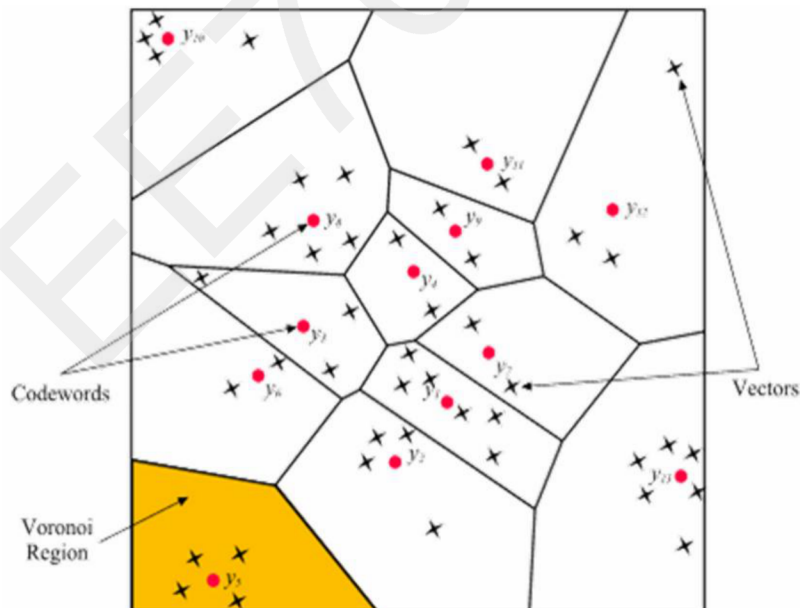
EE708: Fundamentals of Data Science and Machine Intelligence

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Module 5 Vector Quantization (VQ) and Gaussian Mixture Modeling (GMM)

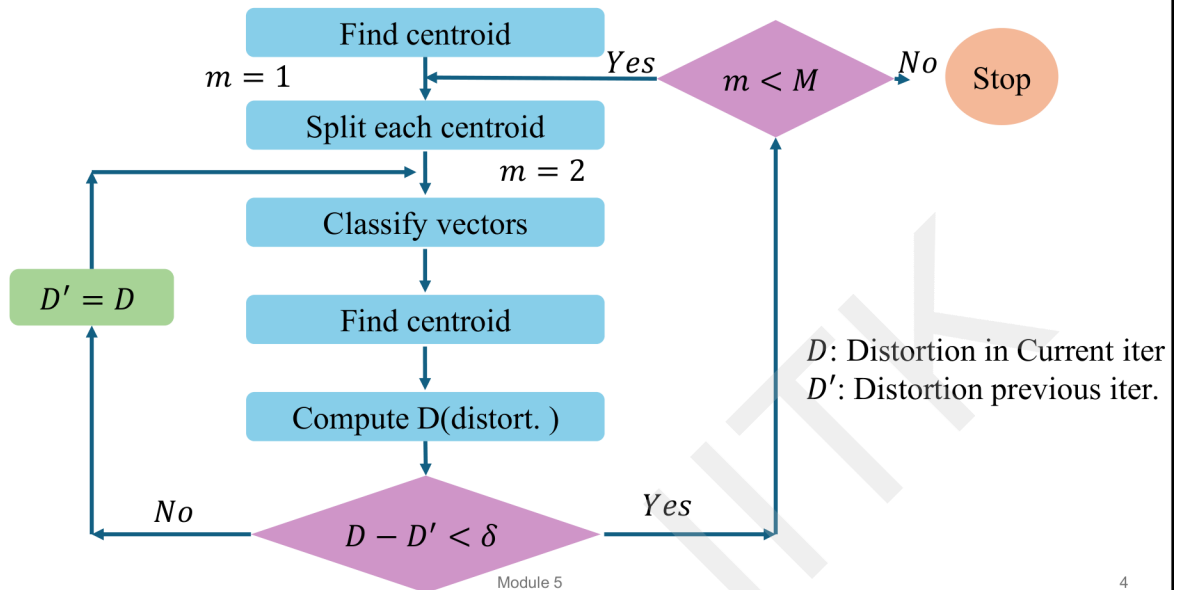
Vector Quantization : Recap from K – Means Algorithm



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Vector Quantization : Algorithm



Vector Quantization : Computing the Centroids

- The average distortion D_i in cell C_i is given by:

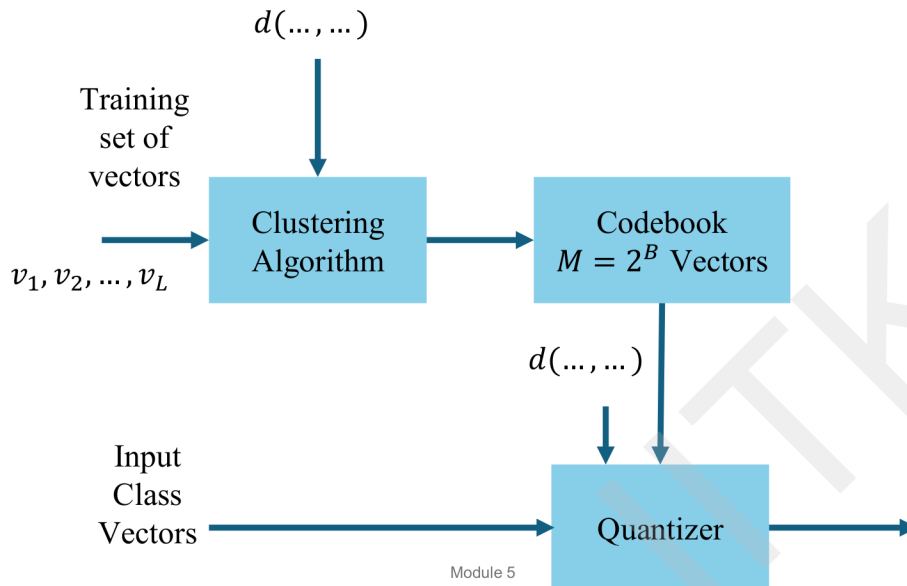
$$D_i = \frac{1}{N} \sum_{x \in C_i} d(x, z_i)$$

where:

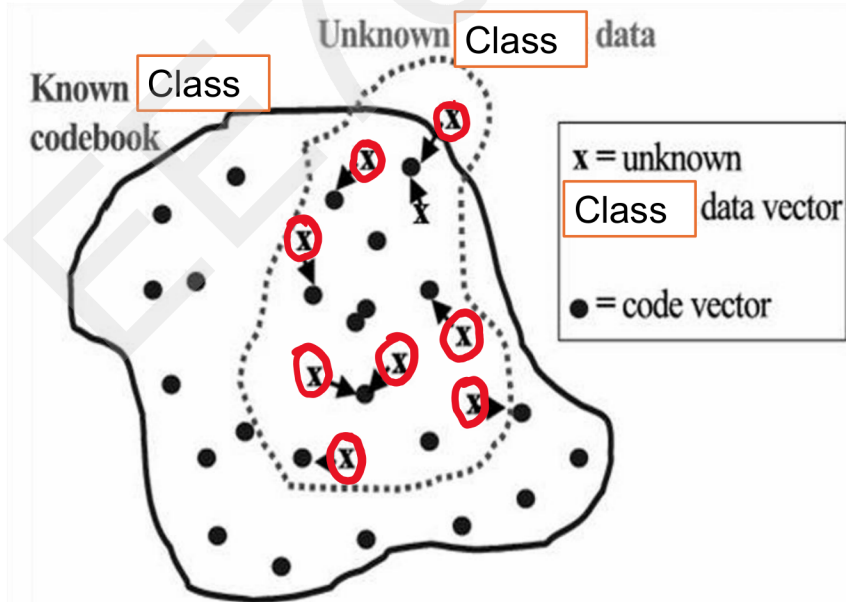
- z_i is the centroid of cell C_i ,
- $d(x, z_i) = (x - z_i)^T (x - z_i)$,
- N is the number of vectors

The centroids that are obtained finally are then stored in a codebook called the VQ codebook.

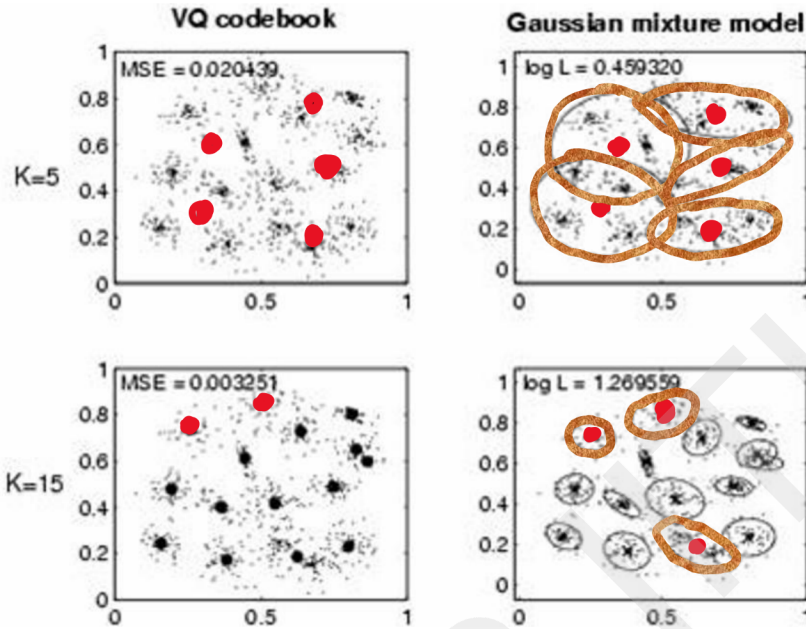
Vector Quantization : Flow Diagram



Decision Making with Vector Quantization : Class Identification



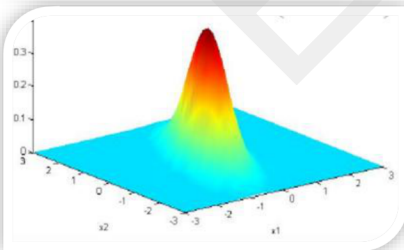
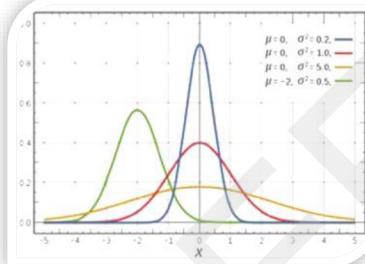
Rubbing shoulders with Gaussian Mixture Modeling (GMM) from VQ



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GMM : Univariate and Multivariate Gaussian Distribution



$$\mu = (0, 0)^T$$

$$\Sigma = \begin{bmatrix} 0.25 & 0.30 \\ 0.30 & 1.00 \end{bmatrix}$$

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- Gaussian distribution for a single variable x :

$$\mathcal{N}(x|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{1/2}} \exp\left\{-\frac{1}{2\sigma^2}(x - \mu)^2\right\}$$

where μ is the mean and σ^2 is the variance.

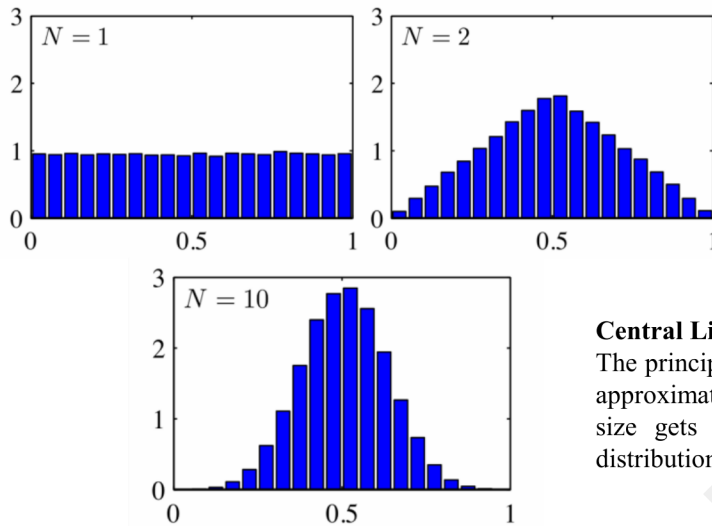
- Multivariate Gaussian for a ' D '-dimensional vector x :

$$\mathcal{N}(x|\mu, \Sigma) = \frac{1}{(2\pi)^{D/2}} \frac{1}{|\Sigma|^{1/2}} \exp\left\{-\frac{1}{2}(x - \mu)^T \Sigma^{-1}(x - \mu)\right\}$$

where:

- μ is a D -dimensional mean vector,
- Σ is a $D \times D$ covariance matrix, and
- $|\Sigma|$ denotes the determinant of Σ .

Histogram Representation



Central Limit Theorem (CLT)

The principle that the distribution of sample means approximates a normal distribution as the sample size gets larger, regardless of the population's distribution.

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Central Limit Theorem

- Let $X_1, X_2, \dots, X_n$ be a random sample drawn from any distribution with a finite mean μ and variance σ^2 . As $n \rightarrow \infty$, the distribution of:

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

“converges” to the distribution $N(0,1)$. In other words,

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \approx N(0,1)$$

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Central Limit Theorem

1. What is $\frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$?

- $E(\bar{X}) = \mu$ and $Var(\bar{X}) = \sigma^2/n$, that means taking the random variable $\bar{X}$, subtracting its mean, and dividing by its standard deviation. **It's a z-score!**

2. “converge” means “convergence in distribution”

$$\lim_{n \rightarrow \infty} P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z\right) = \Phi(z) \text{ for all } z$$

3. CLT is really useful because it characterizes large samples from any distribution. As long as you have a lot of independent samples (from any distribution), then the distribution of the sample mean is approximately normal.

- Unfortunately, there is no CLT analog for variance.
 - But there is an important special case: When $X_1, X_2, \dots, X_n$ are from a normal distribution. The distribution of the sample variance is related to the *Chi-Square* (χ^2) distribution.

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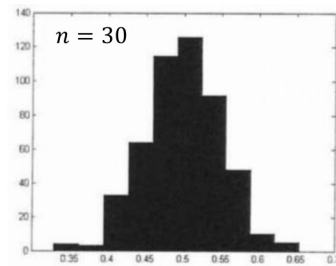
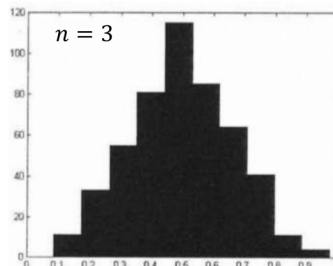
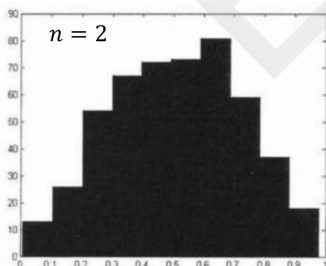
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Central Limit Theorem

Pick large n .

Draw n observations from $U[0,1]$,

	x_1	x_2	x_3	x_n	$\bar{x} = \sum_{i=1}^n x_i / n$
$t = 1$	.21	.76	.57	.84	$(.21 + .76 + \dots) / n$
$\vdots$					
$\vdots$					
$\vdots$					
$t = 1000$					



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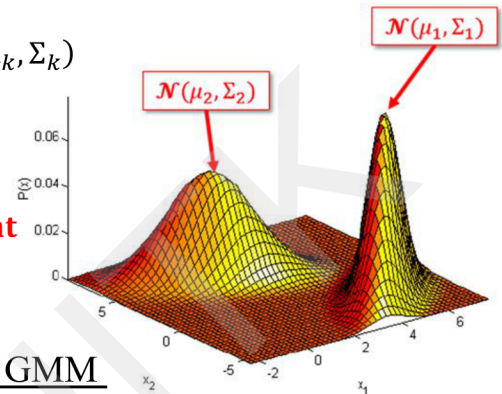
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Gaussian Mixture Model (GMM)

- A density model $p(X)$ may be multi-modal: model it as a mixture of uni-modal distributions (e.g., Gaussians).
- Consider a mixture of K Gaussians:

$$p(X) = \sum_{k=1}^K \pi_k \mathcal{N}(X|\mu_k, \Sigma_k)$$

- π_k : Mixing proportion
- $\mathcal{N}(X|\mu_k, \Sigma_k)$: Mixture component
- $\pi_k \in (0,1)$, μ_k, Σ_k needs to be **Computed/Learnt**



Demonstration of GMM

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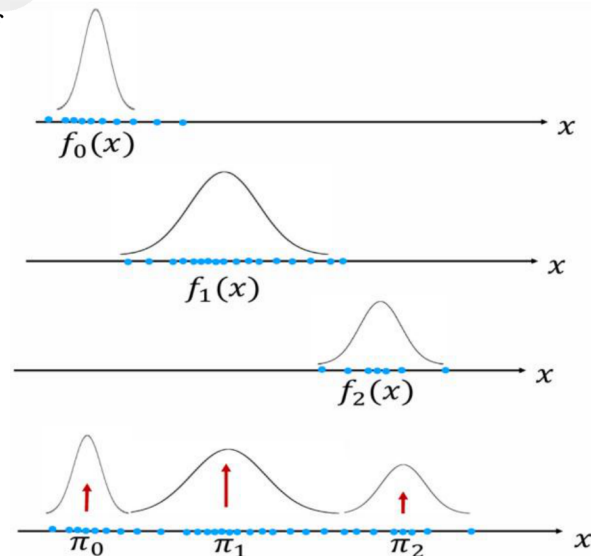
Mixture of Gaussians (MoG)

- Mixture Model is the weighted sum of number of pdfs.
- Weights are determined by distribution π :

$$p(x) = \pi_0 f_0(x) + \pi_1 f_1(x) + \dots + \pi_k f_k(x)$$

$$p(x) = \sum_{i=0}^k \pi_i f_i(x)$$

where $\sum_{i=0}^k \pi_i = 1$



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Mixture of Gaussians (MoG)

- Complex probabilities can be approximated with a linear superposition of K Gaussian densities:

$$p(x) = \sum_{k=1}^K \pi_k \mathcal{N}(x|\mu_k, \Sigma_k)$$

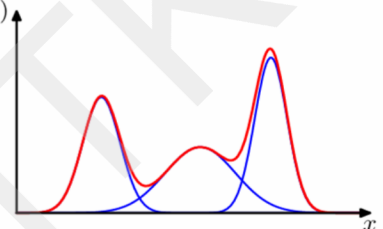
- Define $\mathbf{z} = \{z_1, z_2, \dots, z_K\}$, where $z_k \in \{0,1\}$ and $\sum_k z_k = 1$.

$$p(x, \mathbf{z}) = p(\mathbf{z})p(x|\mathbf{z}) \quad \text{and} \quad p(x) = \sum_{\mathbf{z}} p(\mathbf{z})p(x|\mathbf{z})$$

- $p(z_k = 1) = \pi_k$: ($0 \leq \pi_k \leq 1$) and $\sum_{k=1}^K \pi_k = 1$

- $p(\mathbf{z}) = \prod_{k=1}^K \pi_k^{z_k}$

- $p(x|\mathbf{z}) = \prod_{k=1}^K \mathcal{N}(x|\mu_k, \Sigma_k)^{z_k}$



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Mixture of Gaussians (MoG) : Generative in Nature

- Generative simply means dealing with joint probability $p(x, \mathbf{z})$

$$p(x) = \pi_0 f_0(x) + \pi_1 f_1(x) + \dots + \pi_k f_k(x)$$

- Let's say $f(\cdot)$ is a Gaussian distribution:

$$p(x) = \pi_0 N(x|\mu_0, \sigma_0) + \pi_1 N(x|\mu_1, \sigma_1) + \dots + \pi_k N(x|\mu_k, \sigma_k)$$

$$p(x) = \sum_k N(x|\mu_k, \sigma_k) \pi_k$$

$$p(x) = \sum_k p(x|z_k)p(z_k) \quad z_k \text{ is component } k$$

$$p(x) = \sum_k p(x, z_k)$$

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MoG : Latent Variable Z_k

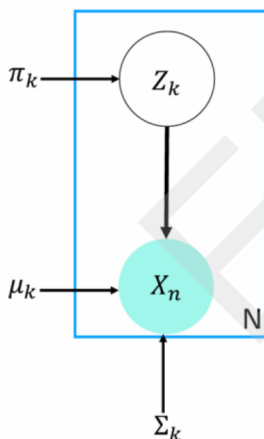
- A variable can be **unobserved (latent)** because:
 - It is an imaginary quantity meant to provide some simplified and abstractive view of the data generation process.
 - e.g., speech recognition models, mixture models (soft clustering)...
 - It is a real-world object and/or phenomenon, but difficult or impossible to measure.
 - e.g., the temperature of a star, causes of a disease, evolutionary ancestors...
 - It is a real-world object and/or phenomenon, but sometimes wasn't measured, because of faulty sensors, etc.
- Discrete latent variables** can be used to partition/cluster data into sub-groups.
- Continuous latent variables (factors)** can be used for dimensionality reduction (factor analysis, etc.).

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MoG Generative Modeling : Latent Variable

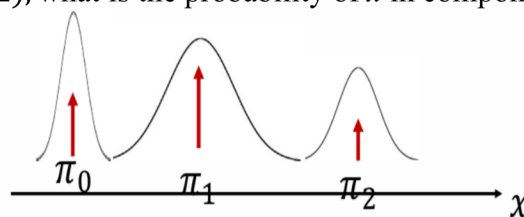


- Z_k is the latent variable represented in a 1-of-K representation.

$$p(z_{nk} | \pi_k) = \prod_{k=1}^K \pi_k^{z_{nk}}$$

$$p(x | z_{nk}, \pi, \mu, \Sigma) = \prod_{k=1}^K (N(x | \mu_k, \Sigma_k))^{z_{nk}}$$

- Given (z, π, μ, Σ) , what is the probability of x in component k



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Finding Probability of Data due to First Component

$$p(x) = \pi_0 \mathcal{N}(X|\mu_0, \sigma_0) + \pi_1 \mathcal{N}(X|\mu_1, \sigma_1) + \pi_2 \mathcal{N}(X|\mu_2, \sigma_2)$$

- Compute the **responsibility of the first component for one data point x** . Let's call that τ_0 :

$$\tau_0 = \frac{\mathcal{N}(X|\mu_0, \sigma_0)\pi_0}{\mathcal{N}(X|\mu_0, \sigma_0)\pi_0 + \mathcal{N}(X|\mu_1, \sigma_1)\pi_1 + \mathcal{N}(X|\mu_2, \sigma_2)\pi_2}$$

$$\tau_0 = \frac{p(x|z_0)p(z_0)}{p(x|z_0)p(z_0) + p(x|z_1)p(z_1) + p(x|z_2)p(z_2)}$$

$$\tau_0 = \frac{p(x, z_0)}{\sum_{k=0}^2 p(x, z_k)} = \frac{p(x, z_0)}{p(x)} = p(z_0|x)$$

- Given a datapoint x , what is the probability of that datapoint in component 0? **$p(0|x)=1$**
- If I have 100 datapoints and 3 components, what is the size of τ ? **100×3**

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Finding Cluster Memberships

- The conditional of $p(z_{nk}|x, \theta)$ (**Posterior**) can be derived using Bayes' rule.
- The responsibility that a mixture component takes for explaining an observation x :

$$\tau(z_k) = p(z_k = 1|x) = \frac{p(z_k = 1)p(x|z_k = 1)}{\sum_{j=1}^K p(z_j = 1)p(x|z_j = 1)}$$

$$\tau(z_k) = \frac{\pi_k \mathcal{N}(x|\mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(x|\mu_j, \Sigma_j)}$$

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Computing the Mixture of Gaussians

- What is the probability of picking a mixture component (Gaussian model) = $p(z) = \pi_i$

AND

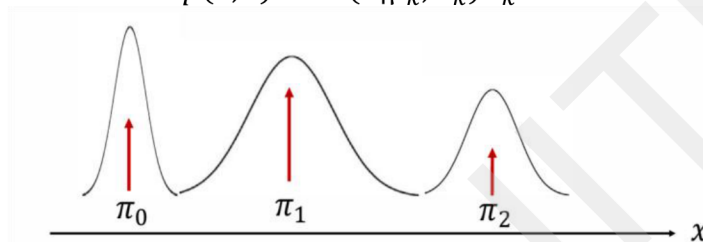
- Picking data from that specific mixture component: $p(x | z)$

z is latent, we observe x , but z is hidden.



$$p(x, z) = p(x|z)p(z) \Rightarrow \text{Generative model, Joint distribution}$$

$$p(x, z) = \mathcal{N}(x|\mu_k, \sigma_k)\pi_k$$



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Computing the Mixture of Gaussians : Parameters to Find ?

What are GMM parameters (Together represented by vector θ) ?

- Mean: μ_k
- Variance: σ_k
- Size: π_k
- Marginal probability distribution:

$$p(x|\theta) = \sum_k p(x, z_k|\theta) = \sum_k p(x|z_k, \theta)p(z_k|\theta) = \sum_k \mathcal{N}(x|\mu_k, \sigma_k)\pi_k$$

where:

- $p(z_k|\theta) = \pi_k$: Select a mixture component with probability π .
- $p(x|z_k, \theta) = \mathcal{N}(x|\mu_k, \sigma_k)$: Sample from that component's Gaussian.

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How to Compute π_k, μ_k, Σ_k ? What should we do???

- We use a method called **Maximum Likelihood Estimation (MLE)** to solve the problem:

$$p(x) = p(x|\theta) = \sum_k p(x, z_k|\theta) = \sum_k p(z_k|\theta)p(x|z_k, \theta) = \sum_{k=0}^K \pi_k \mathcal{N}(x|\mu_k, \Sigma_k)$$

- Identify a likelihood function, why?
- Because we use the likelihood function to optimize the probabilistic model parameters!

$$\arg \max_{\pi, \mu, \Sigma} p(x|\theta) = p(x|\pi, \mu, \Sigma) = \prod_{n=1}^N p(x_n|\theta) = \prod_{n=1}^N \sum_{k=0}^K \pi_k \mathcal{N}(x_n|\mu_k, \Sigma_k)$$

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Compute π_k, μ_k, Σ_k

$$\arg \max_{\pi, \mu, \Sigma} p(x) = p(x|\pi, \mu, \Sigma) = \prod_{n=1}^N p(x_n|\theta) = \prod_{n=1}^N \sum_{k=0}^K \pi_k \mathcal{N}(x_n|\mu_k, \Sigma_k)$$

$$\ln[p(x)] = \ln[p(x|\pi, \mu, \Sigma)]$$

- As usual: Identify a likelihood function

$$\ln p(x|\pi, \mu, \Sigma) = \sum_{n=1}^N \ln \left\{ \sum_{k=0}^K \pi_k \mathcal{N}(x_n|\mu_k, \Sigma_k) \right\}$$

- And set partials to zero

$\mathcal{N}(\cdot) =$ Multivariate Gaussian

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Maximum Likelihood Estimation of GMM Parameters

Estimation of Means:

μ_k

$$\ln p(x|\pi, \mu, \Sigma) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k N(x_n | \mu_k, \Sigma_k) \right\}$$

$$\frac{\partial \ln p(x|\pi, \mu, \Sigma)}{\partial \mu_k} = \sum_{n=1}^N \frac{\pi_k N(x_n | \mu_k, \Sigma_k)}{\sum_j \pi_j N(x_n | \mu_j, \Sigma_j)} \Sigma_k (x_n - \mu_k) = 0$$

$$= \sum_{n=1}^N \tau(z_{nk}) \Sigma_k (x_n - \mu_k) = 0$$

$$\mu_k = \frac{\sum_{n=1}^N \tau(z_{nk}) x_n}{\sum_{n=1}^N \tau(z_{nk})}$$

mult. both sides by Σ_k^{-1}
assume non-sing.

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Maximum Likelihood Estimation of GMM Parameters

Estimation of Covariance:

$$\ln p(x|\pi, \mu, \Sigma) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k N(x_n | \mu_k, \Sigma_k) \right\}$$

$$\Sigma_k = \frac{1}{\sum_{n=1}^N \tau(z_{nk})} \sum_{n=1}^N \tau(z_{nk}) (x_n - \mu_k)(x_n - \mu_k)^T$$

Derivation involved

Sec. 2.3.4 : Bishop

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Maximum Likelihood Estimation of GMM Parameters

Estimation of Mixing Term:

$$\ln p(x|\pi, \mu, \Sigma) + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right)$$

$$0 = \sum_{n=1}^N \frac{N(x_n|\mu_k, \Sigma_k)}{\sum_j \pi_j N(x_n|\mu_j, \Sigma_j)} + \lambda$$

$$\pi_k = \frac{\sum_{n=1}^N \tau(z_{nk})}{N}$$

method of
Lagrange
Multipliers
to take
care of
constraint
 $\sum_K \pi_k = 1$

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Maximum Likelihood Estimation of GMM Parameters

$$\mu_k = \frac{\sum_{n=1}^N \tau(z_{nk}) x_n}{N_k}$$

$$\Sigma_k = \frac{1}{N_k} \sum_{n=1}^N \tau(z_{nk}) (x_n - \mu_k)(x_n - \mu_k)^T$$

$$\pi_k = \frac{N_k}{N}, \quad N_k = \sum_{n=1}^N \tau(z_{nk})$$

Not a closed form solution!!

τ is not known exactly.

What next?

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Expectation Maximization (EM) for Gaussian Mixtures (MOG)

- The steps involved in maximizing the likelihood with respect to the model parameters:
 1. Initialize the means μ_k , covariances Σ_k , and mixing coefficients π_k , and evaluate the log likelihood's initial value.
 2. **E Step:**

$$\gamma(z_{nk}) = \frac{\pi_k N(x_n | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j N(x_n | \mu_j, \Sigma_j)}$$

3. **M Step:** Re-estimate the model parameters:

$$\mu_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^N \tau(z_{nk}) x_n$$

$$\Sigma_k^{\text{new}} = \frac{1}{N_k} \sum_{n=1}^N \tau(z_{nk}) (x_n - \mu_k^{\text{new}})(x_n - \mu_k^{\text{new}})^T$$

where
 $\pi_k^{\text{new}} = \frac{N_k}{N}, \quad N_k = \sum_{n=1}^N \gamma(z_{nk})$

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Expectation Maximization (EM) for Gaussian Mixtures (MOG)

4. **Evaluate the log-likelihood:**

$$\ln(p(X | \mu, \Sigma, \pi)) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k N(x_n | \mu_k, \Sigma_k) \right\}$$

Check for convergence of parameters or log-likelihood.

If the criterion fails, repeat steps 2, 3, and 4.

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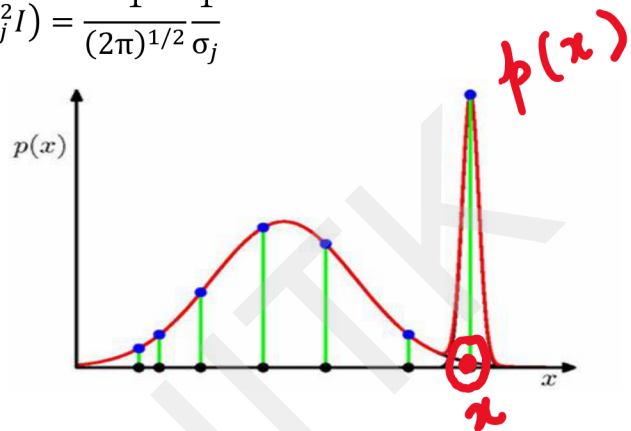
Singularity Problem

- Consider a data point's mean $\mu_j = x_n$ for some value of n :

$$\mathcal{N}(x_n | x_n, \sigma_j^2 I) = \frac{1}{(2\pi)^{1/2}} \frac{1}{\sigma_j}$$

where I is a unit matrix.

- If $\sigma_j \rightarrow 0$, then LHS goes to infinity.

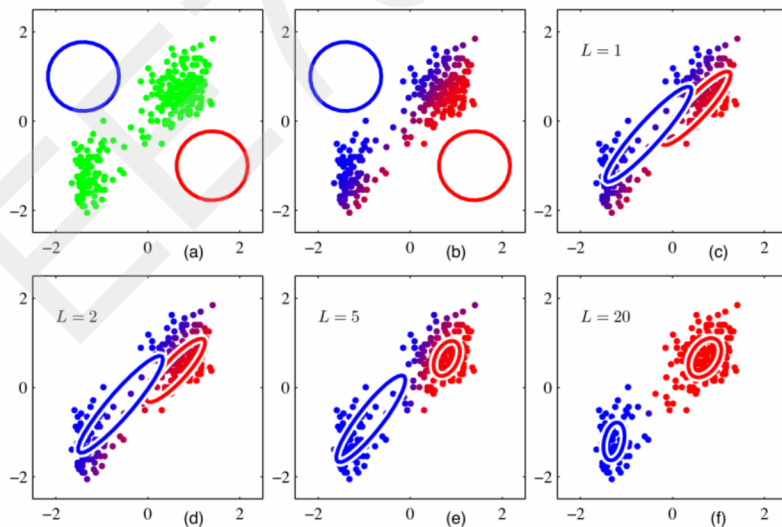


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Illustration of EM and Convergence

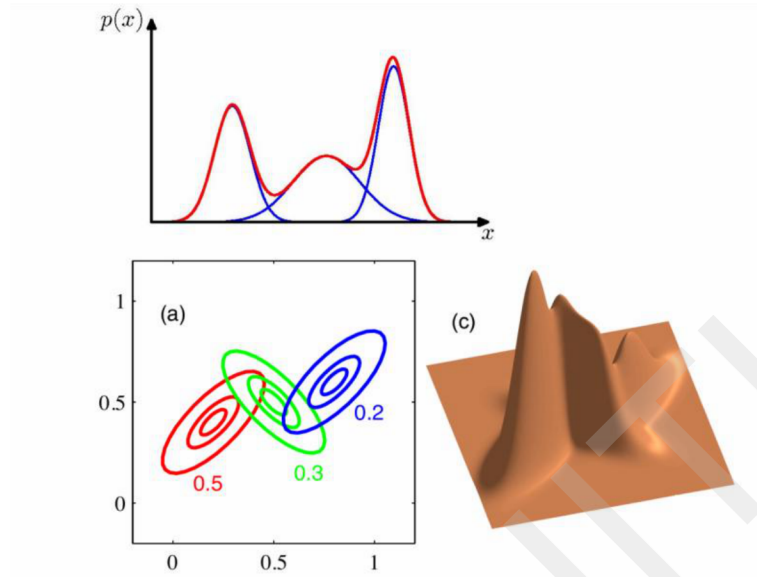


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Illustration of a GMM with 3 Mixtures



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Generalized EM Algorithm

- Steps for maximizing the likelihood function $p(X|\theta)$ with respect to θ

1. Choose an initial value for parameters θ^{old} .

2. **E Step:** Evaluate $p(Z|X, \theta^{old})$.

3. **M Step:** Evaluate θ^{new} by:

$$\theta^{new} = \arg \max_{\theta} Q(\theta, \theta^{old})$$

where: $Q(\theta, \theta^{old}) = \sum_z p(z|X, \theta^{old}) \ln[p(X, Z|\theta)]$ ↖ KLD

4. Check for convergence of log-likelihood or parameter values. If the criterion fails:

$$\theta^{old} \leftarrow \theta^{new}$$

Return to Step 2.

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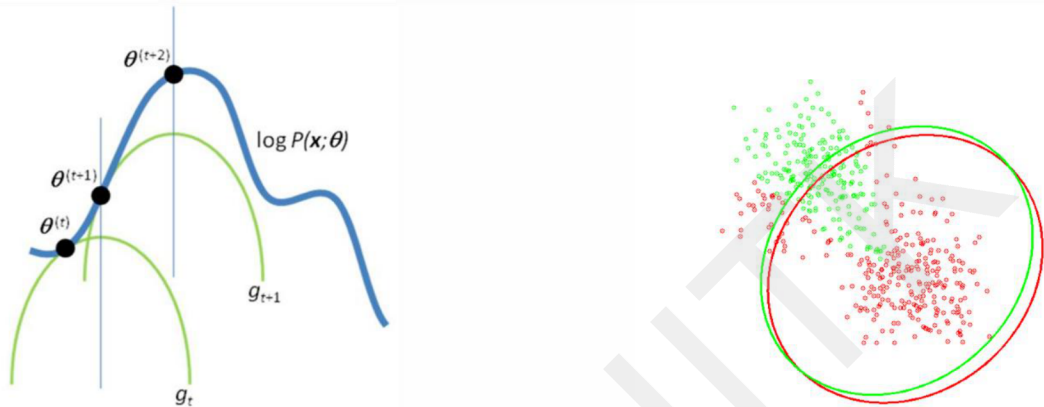
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Expectation Maximization

- The EM algorithm improves on the GMM parameter estimates by iteratively checking for the condition

$$p(X|\lambda^{k+1}) > p(X|\lambda^k)$$



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Summary

- K-means makes hard decisions.
 - Each data point gets assigned to a single cluster.
- GMM/EM makes soft decisions.
 - Each data point can yield a posterior $p(z|x)$.
- K-means is a special case of EM.

Demo link: <https://lukapopijac.github.io/gaussian-mixture-model>

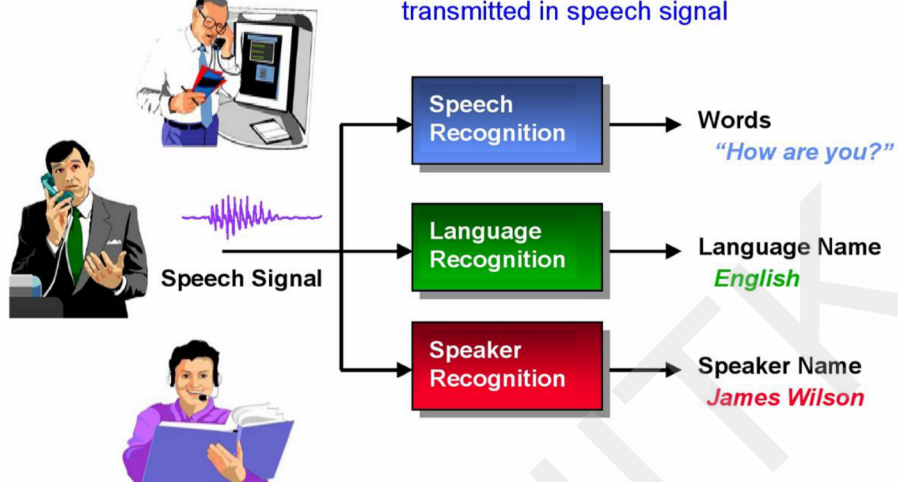
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Case Study: Speaker Recognition and Verification

Goal: Automatically extract information transmitted in speech signal



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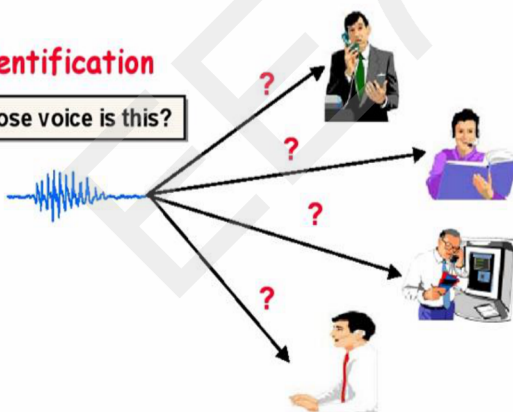
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Speaker Identification versus Verification

Identification

Whose voice is this?



Verification/Authentication

Is this Bob's voice?



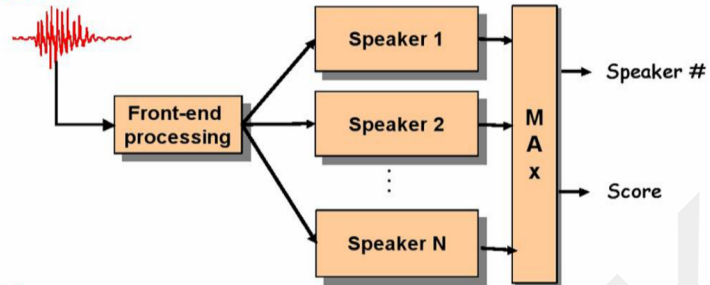
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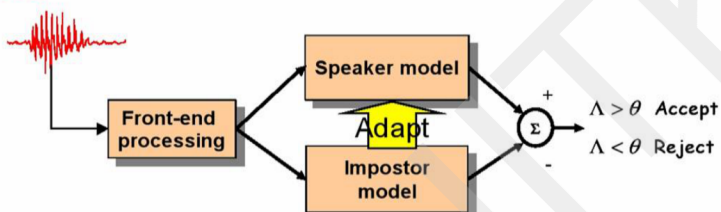
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Speaker Identification versus Verification

Identification



Verification

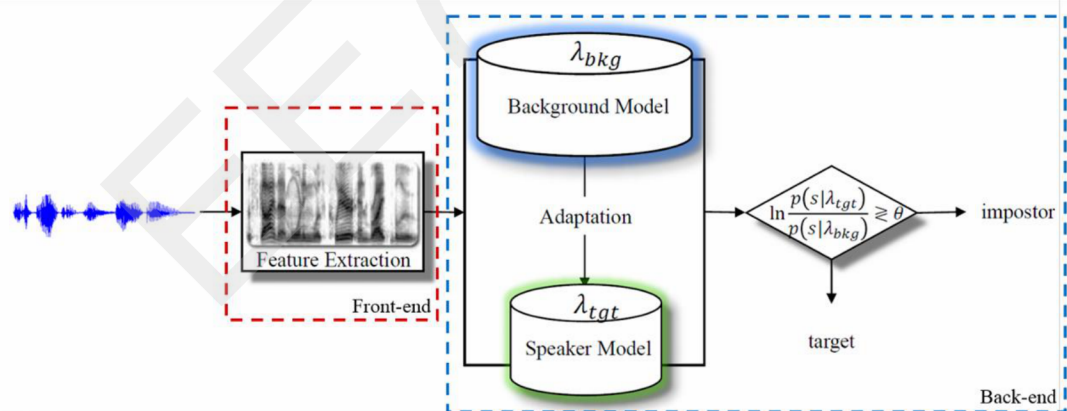


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Speaker Verification using Background Model



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Training the Speaker Models

- Clusters are formed within the training data
- Each cluster is then represented with multiple Gaussian pdfs.
- The union of many such Gaussian pdfs is a GMM
- The most common approach to estimate the GMM parameters is the maximum likelihood (ML) estimation
- $p(X|\lambda)$ is maximized with respect to λ .
- $p(X|\lambda)$ is the conditional probability and vector $X = \{x_0, x_1, \dots, x_{M-1}\}$ is the set of all feature vectors belonging to a particular speaker
- No closed form solution to the maximum likelihood estimation
- Convergence is guaranteed only when large enough data is available

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Decision making for Speaker ID

- Let the number of models representing different speakers be S .
- Hence λ_j , where $j = \{1, 2, 3, \dots, S\}$, is the set of GMMs under consideration.
- For each test speaker, feature vectors x_n at time n are extracted.
- The probability of each model given the feature vectors x_n is given by

$$P(\lambda_j|x_n) = \frac{p(x_n|\lambda_j)P(\lambda_j)}{P(x_n)}$$
- Since $P(x_n)$ is a constant and $P(\lambda_j)$, the a priori probabilities, are assumed to be equal, the problem is reduced to finding the λ_j that maximizes $p(x_n|\lambda_j)$.

- But $p(x_n|\lambda_j)$ is given by

$$P(x_n|\lambda_j) = p(\{x_0, x_1, \dots, x_{M-1}\}|\lambda_j)$$

where M is the number of feature vectors belonging to a particular speaker.

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Module 5

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Decision making for Speaker ID

- Assuming that each speaker is statistically independent,

$$p(\{x_0, x_1, \dots, x_{M-1}\}|\lambda_j) = \prod_{m=0}^{M-1} p(x_m|\lambda_j)$$

- Applying logarithm on above equation and simplifying for S

$$S_r = \max_{1 \leq j \leq S} \sum_{m=0}^{M-1} \log[p(x_m|\lambda_j)]$$

where S_r is declared as the speaker to which the feature vectors belong.

Note: $\{S_r, r = \{1, 2, 3, \dots, S\}\}$ is the set of all speakers.